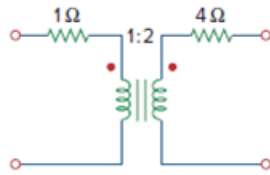


Problem

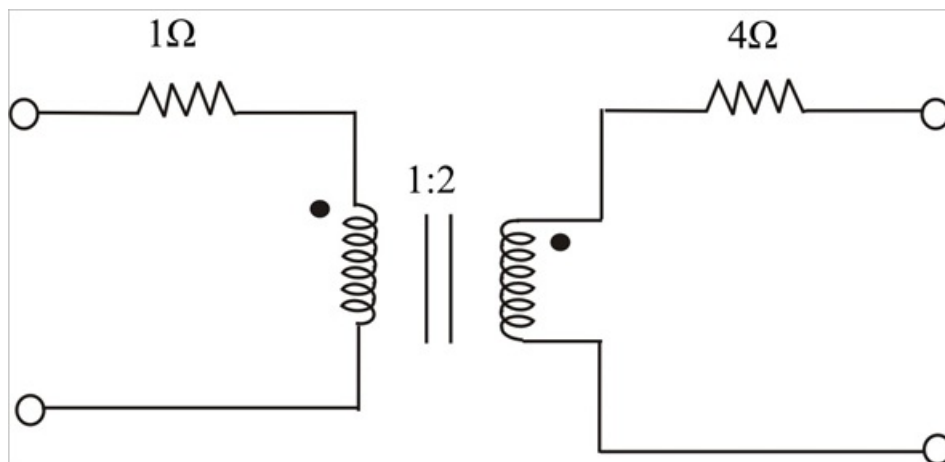
Determine the h parameters for the network in Fig. 19.93.

Figure 19.93



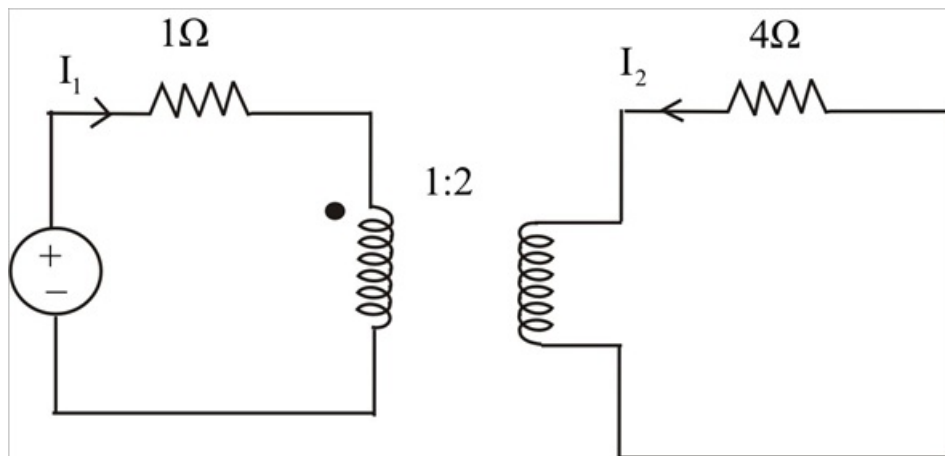
Step-by-step solution

Step 1 of 6



Step 2 of 6

$V_2 = 0$ The circuit becomes

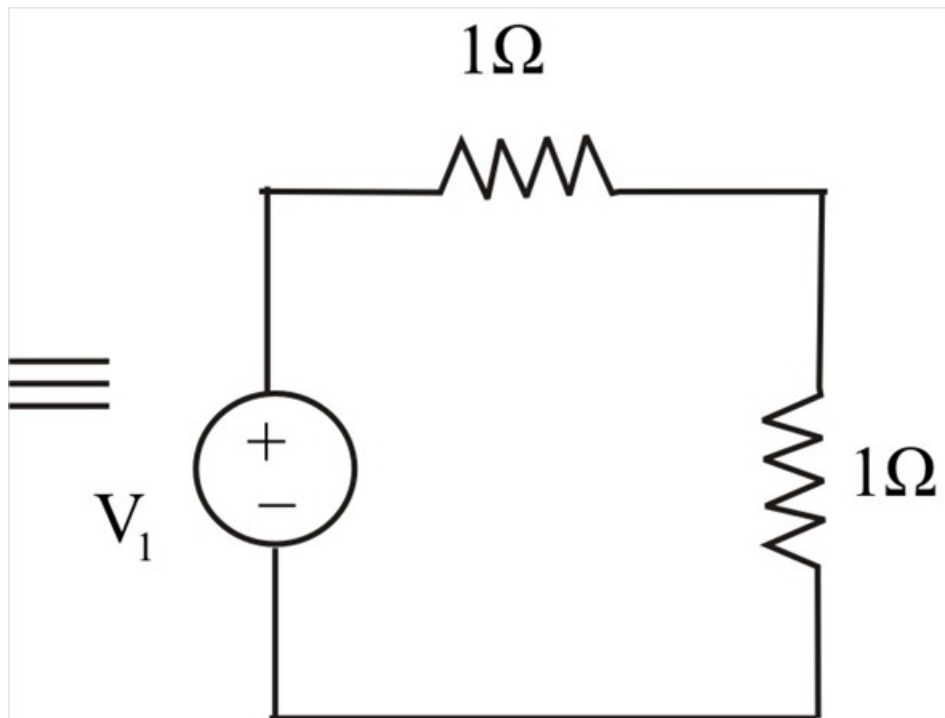


Step 3 of 6

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$

$$R_1' = \frac{R_2'}{n^2} \Rightarrow R_1' = \frac{4}{4} = 1\Omega$$



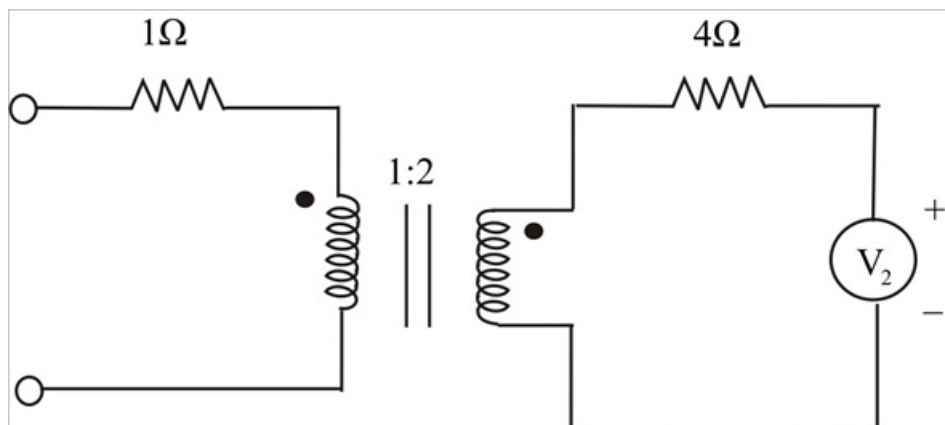
Step 4 of 6

$$\frac{V_1}{I_1} = \frac{V_1}{\frac{V_1}{2}} = 2\Omega$$

For trans former $I_1 = +nI_2 \Rightarrow \frac{I_2}{I_1} = \frac{1}{n} = \frac{1}{2} = 0.5$

Step 5 of 6

For $I_1 = 0$



Step 6 of 6

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} = \frac{V_1}{V_2} = -\frac{1}{n} = -\frac{1}{2} = -0.5$$

[$\therefore$ Due to the direction of current since]

$$I_1 = 0 \Rightarrow h_{22} = 0$$

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$

As the resistance seen on the out put side of the transformer is ∞

$$h = \begin{bmatrix} 2\Omega & 0.5 \\ -0.5 & 0 \end{bmatrix}$$